

Minjae Kwon

hbt9su@virginia.edu • Charlottesville, VA • Website • Google Scholar

Education

University of Virginia , Charlottesville, VA Ph.D. Candidate, Computer Science Dissertation: <i>Safe Decision-Making When the World Changes</i>	Aug 2022 – Present (Expected May 2027)
Kyungpook National University , Daegu, South Korea Ph.D. in Mathematics Dissertation: <i>On a Generalized Class of Noetherian Rings</i>	Mar 2017 – Aug 2022
Kyungpook National University , Daegu, South Korea B.S. and M.S. in Mathematics Thesis: <i>The Krull Dimension of Composite Power Series Rings over Valuation Rings</i>	Mar 2011 – Feb 2017

Research Experience

University of Virginia , Charlottesville, VA <i>Graduate Research Assistant</i> , Department of Computer Science Advisor: Lu Feng	Aug 2022 – Present
- Research on safe and generalizable reinforcement learning, working on runtime safety, in-context learning, and deployment-time adaptation under distribution shift without extra retraining, with applications to simulated health-care and robotics. - Designed reward functions for reinforcement learning that incentivize partial completion of tasks specified in Linear Temporal Logic, addressing sparse-reward limitations in uncertain environments with applications to complex robotic tasks.	
University of Toronto , Toronto, Canada <i>Visiting Scholar</i> (Mitacs Globalink Research Award) Host: Chi-Guhn Lee	Sep 2021 – Feb 2022
Developed a recursive time-series data augmentation method to address data scarcity, and proved bounds on the augmented model's expected error.	
Kyungpook National University , Daegu, South Korea <i>Graduate Researcher</i> , Department of Mathematics Advisor: Jung Wook Lim	Mar 2015 – Jul 2022
Research on ring theory and commutative algebra, including generalizations of Noetherian rings, Krull dimension of power series rings, and star operations on fractional ideals.	

Peer-Reviewed Publications

* denotes equal contribution.

- [1] **Minjae Kwon**^{*}, Amir Moeini^{*}, Shangdong Zhang, Lu Feng. *Latent Q-Barrier Shielding for Safe In-Context Reinforcement Learning*. Conference on Neural Information Processing Systems (NeurIPS), 2026. To appear.
- [2] Amir Moeini^{*}, **Minjae Kwon**^{*}, Alper Kamil Bozkurt, Yuichi Motai, Rohan Chandra, Lu Feng, Shangdong Zhang. *Safe In-Context Reinforcement Learning*. International Conference on Machine Learning (ICML), 2026.
- [3] **Minjae Kwon**, Josephine Lamp, Lu Feng. *Safety Generalization Under Distribution Shift in Safe Reinforcement Learning: A Diabetes Testbed*. International Conference on Machine Learning (ICML), 2026.
- [4] **Minjae Kwon**, Ingy ElSayed-Aly, Lu Feng. *Adaptive Reward Design for Reinforcement Learning*. Conference on Uncertainty in Artificial Intelligence (UAI), 2025.
- [5] Amine Mohamed Aboussalah, **Minjae Kwon**, Raj G Patel, Cheng Chi, Chi-Guhn Lee. *Recursive Time Series*

Data Augmentation. International Conference on Learning Representations (ICLR), 2023.

[6] **Minjae Kwon**, Jung Wook Lim. *On Nonnil- S -Noetherian Rings*. Mathematics, 8(9), 1428, 2020.

Preprints and Papers Under Review

[1] **Minjae Kwon**^{*}, Amir Moeini^{*}, Shangdong Zhang, Lu Feng. *Algorithm Distillation for Safe In-Context Reinforcement Learning*. Under review, 2026.

[2] **Minjae Kwon**, Tyler Ingebrand, Ufuk Topcu, Lu Feng. *Adaptive Shielding for Safe Reinforcement Learning under Hidden-Parameter Dynamics Shifts*. Preprint, 2026.

Research Award

Mitacs Globalink Research Award, National Research Foundation of Korea (NRF) and Mitacs 2021
Nationwide Korea–Canada graduate research award with 20 Korea-to-Canada awards. Selected based on a research proposal on reinforcement learning and time-series modeling for financial applications. Award value: C\$20,000.

Talks

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| [1] “Safe Decision-Making When the World Changes”
Korea Institute for Advanced Study, Seoul, South Korea | Jul 2026 |
| [2] “Introduction to Deep Reinforcement Learning”
Open Seminar on Algebras, Combinatorics and Algorithms, Virtual | Aug 2022 |
| [3] “Introduction to Mathematics for Artificial Intelligence” (outreach)
Pohang Jecheol High School, Pohang, South Korea | Jul 2021 |
| [4] “Deep Reinforcement Learning for Trading”
The Korea Society for Computational and Applied Mathematics, Daejeon, South Korea | Aug 2020 |
| [5] “Some Global Dimension of Power Series Rings over w -Coherent Rings”
Mediterranean Conference on Pure and Applied Mathematics, Paris, France | Aug 2019 |
| [6] “Basic Properties of w -SFT-Rings”
Joint Meeting of the Korean Mathematical Society and the German Mathematical Society, Seoul, South Korea | Oct 2018 |
| [7] “Some Results on w -SFT-Rings”
The 10th KOOK-TAPU Joint Seminar on Knots and Related Topics, Busan, South Korea | Jul 2018 |
| [8] “Note on w -SFT Rings”
Honam–YOUNGNAME Mathematical Societies Meeting, Jeju, South Korea
The Korea Society for Computational and Applied Mathematics, Daejeon, South Korea | Jun 2018 |
| [9] “Note on Nonnil- S -Noetherian Rings”
The Korean Mathematical Society Spring Meeting, Seoul, South Korea | Apr 2018 |
| [10] “The Krull Dimension of Composite Power Series Rings over Valuation Rings”
The 50th Symposium on Ring Theory and Representation Theory, Yamanashi, Japan
The 9th KOOK-TAPU Joint Seminar on Knots and Related Topics, Osaka, Japan | Jul, Oct 2017 |
| [11] “On Nonnil- S -Noetherian Rings”
The Korean Mathematical Society Spring Meeting, Suwon, South Korea
Joint Mathematics Meetings, Seattle, United States | Jan, Apr 2016 |

Teaching and Mentoring Experience

Teaching Experience

Teaching Assistant, University of Virginia

- Cyber-Physical Systems: Formal Methods, Safety & Security · CS 6763/SYS 6763 Spring 2024, Spring 2025
Mentored final projects in explainable AI and safe reinforcement learning
- Artificial Intelligence · CS 4710 Fall 2023, Fall 2025
Developed a Google Colab workflow for course assignments

Teaching Assistant, Kyungpook National University

- Modern Algebra I & Seminar · MATH413 Spring 2015, Spring 2016, Spring 2020
Led lab sessions for approximately 30–40 students and co-designed midterm and final exams
- Modern Algebra II & Seminar · MATH414 Fall 2015, Fall 2016
Led lab sessions for approximately 20–25 students and co-designed midterm and final exams
- Modern Algebra · MTED334 Fall 2017
- Applied Mathematics · MATH401 Spring 2017
- Introduction to Graph Theory · MATH420 Spring 2017
- Set Theory & Mathematical Logic · MATH292 Spring 2017, Fall 2017
- Number Theory · MATH235 Fall 2017
- Calculus I · CLTR211 Summer 2017, Winter 2017, Summer 2018, Spring 2020, Spring 2021
- Calculus II · CLTR212 Winter 2018, Winter 2019
- Linear Algebra I · MATH231 Spring 2018
- Special Topics in Geometry · MATH429 Fall 2019
Co-designed lecture slides

Guest Lectures, Kyungpook National University

- Modern Algebra I & Seminar · MATH413 Spring 2015, Spring 2016, Spring 2020
Groups and operations, approximately 30–40 students
- Modern Algebra II & Seminar · MATH414 Fall 2015, Fall 2016
Rings and fields, approximately 20–25 students
- Linear Algebra I · MATH231 Spring 2018
Matrix algebra and linear maps, 42 students
- Set Theory & Mathematical Logic · MATH292 Spring 2017, Fall 2017
Propositional logic and set operations, 42 students
- Number Theory · MATH235 Fall 2017
Modular arithmetic and prime numbers, 36 students
- Calculus I · CLTR211 Summer 2017, Winter 2017, Summer 2018, Spring 2020, Spring 2021
Infinite series and convergence tests, approximately 50-70 students
- Calculus II · CLTR212 Winter 2018, Winter 2019
Double integrals, vector fields, Green's theorem, and Stokes' theorem, approximately 40-50 students

Tutor, Kyungpook National University

- Math Help Center Fall 2015, Winter 2015, Spring 2016, Fall 2016, Spring 2018, Fall 2018
Tutored students in Introductory Calculus and Calculus I–II in coordination with teaching faculty

Mentoring Experience

Undergraduate Research Mentor, University of Virginia

Fall 2026

- *Mentoring two undergraduate students on safe reinforcement learning for personalized diabetes management using physiological simulators.*

Charlottesville High School Mentoring Program

Fall 2026

- *Mentor Charlottesville High School students through biweekly school visits, helping students develop and refine their own project ideas. Currently advising a project on analyzing U.S. housing and mortgage data.*

Externship Mentor, Ardsley High School

May–Jun 2026

- *Mentored a high school student through a 30-day externship program, introducing artificial intelligence and reinforcement learning and guiding the development of an RL agent for an Atari environment.*

Gyunggu High School Mentoring Program

Spring 2012, Fall 2012

- *Mentored four high school students through home visits, teaching high school mathematics and providing broader academic guidance.*

Awards, Honors, and Scholarships

Awards and Honors

- *Gold Reviewer Award*, International Conference on Machine Learning (ICML) 2026
- *Award for Presentation Excellence*, The Korea Society for Computational and Applied Mathematics 2019
- *Outstanding Participation Award*, Brain Korea 21 Plus 2017
- *Best Tutor Award*, Math Help Center, Kyungpook National University 2016
- *Award for Presentation Excellence*, TAPU (Graduate Student Workshop on Mathematics) 2016
- *Silver Medal*, 33rd Mathematics Competition for University Students, Korean Mathematical Society 2014

Scholarships

- *Conference Scholarship*, Conference on Uncertainty in Artificial Intelligence (UAI) 2025
- *Brain Korea 21 Research Scholarship*, Kyungpook National University 2015–2021
- *Graduate Scholarship* (admission with highest distinction), Kyungpook National University 2015
- *Scholarship for Academic Excellence*, Kyungpook National University 2014
- *Academic Support Scholarship*, Government of the Republic of Korea 2014
- *Scholarship for Academic Excellence*, Kyungpook National University 2012
- *Vision Scholarship*, Government of the Republic of Korea 2012
- *National Excellence Dream Scholarship*, Government of the Republic of Korea 2011
- *Alumni Scholarship*, Department of Mathematics, Kyungpook National University 2011

Professional Activities

Conference Reviewing

- AAAI Conference on Artificial Intelligence (AAAI) 2026
- Conference on Neural Information Processing Systems (NeurIPS) 2026
- International Conference on Machine Learning (ICML) 2026
- International Conference on Learning Representations (ICLR) 2025–2026
- Conference on Uncertainty in Artificial Intelligence (UAI) 2026
- Reinforcement Learning Conference (RLC) 2026
- IEEE International Conference on Robotics and Automation (ICRA) 2026
- Hybrid Systems: Computation and Control / International Conference on Cyber-Physical Systems 2026

Journal Reviewing

- Transactions on Machine Learning Research (TMLR) 2026
- Filomat 2026
- Rocky Mountain Journal of Mathematics 2026
- Electronics and Telecommunications Research Institute (ETRI) Journal 2026
- ACM Transactions on Cyber-Physical Systems (TCPS) 2024–2026

Technical Skills

- **Programming:** Python, C/C++
- **Scientific computing:** PyTorch, NumPy, SciPy, TensorFlow/Keras, JAX
- **Other:** L^AT_EX, Git, Linux